Coupled Mechanics — Calculus Edition

Field 23 of 80 — The LKVexa Compendium carried to the continuum

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Coupled Mechanics — Calculus Image

Coupling is off-diagonal blocks. Every entry here is a statement about a block matrix $\left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ in which B and C carry the physics: symmetric off-diagonals give a variational structure and a symmetric solver, antisymmetric ones give gyroscopic behaviour, and the relative size of BC to AD is the coupling strength. Onsager reciprocity is the statement that $B = C^{\top}$.

Each entry carries its originating equation from the Linear Algebra edition, its 8S image, its calculus image under $\mathsf{T}_{\mathrm{cal}}$, a six-step proof, and the original discussion. The schema (Q1)–(Q6), the master theorems Θ-A…Θ-J and the lemmas D1…D14 are established in `00_Foundations_Calculus_Transform.md`.

Thermomechanical Coupling

1. Thermal Strain

Origin (LA edition).

$$\varepsilon^{\mathrm{th}} = \alpha\, \Delta T\, I$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(1)} = \mu_k^{(1)} \phi_k^{(1)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(1)} \le \mu_2^{(1)} \le \cdots \le \mu_N^{(1)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(1)}) = \mu_k^{(1)} \phi_k^{(1)}, \qquad \mu_2^{(1)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_1 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(1)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_1$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A scalar multiple of the identity for isotropic materials.

Structure: Purely volumetric with zero deviatoric part, so free thermal expansion produces no shear and no stress. Anisotropic materials have a symmetric expansion tensor α whose eigenvectors are the principal expansion directions.

Reduction: Thermal strain is a scalar times the identity.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

2. Duhamel–Neumann Constitutive Law

Origin (LA edition).

$$\sigma = C : (\varepsilon - \alpha \Delta T \, I)$$

Input (8S edition).

$$y_2(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_2)\Big] z(u), \qquad z_2(u) = \begin{bmatrix} c_2 \\ r_2 u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_2)(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_2\, \psi_2(c')\, p(c')\, dc', \qquad \partial_\tau \psi_2 = -\mathcal{L}_\star \psi_2, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_2 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_2)$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(2)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_2$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Hooke's law with a thermal eigenstrain subtracted.

Structure: Linear in both ε and ΔT, so the stress is $C\varepsilon - \beta\Delta T$ with $\beta = C : \alpha I$ the thermal stress coefficient. In block form the mechanical–thermal coupling appears as the column $-\beta$.

Reduction: Coupling is a linear term in the temperature.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

3. Thermal Stress in Full Constraint

Origin (LA edition).

$$\sigma = -\frac{E\alpha\Delta T}{1-\nu}$$

Input (8S edition).

$$\max_{n_3 \in \mathbb{Z}^5} \; \Delta_{8S} = \text{Score}(\mathcal{M}_8) - \text{Score}(\mathcal{M}_7) \quad \text{s.t.} \quad \|n_3\|_1 \leq M, \;\; \eta^{(3)}_{\text{ind}} \geq \varepsilon_{\text{ind}}$$

Calculus image.

$$\delta\Big[\Delta_{8S} - \lambda(\|n_3\|_1 - M) - \nu(\varepsilon_{\text{ind}} - \eta^{(3)}_{\text{ind}})\Big] = 0, \qquad \lambda, \nu \geq 0, \qquad \lambda(\|n_3\|_1 - M) = \nu(\varepsilon_{\text{ind}} - \eta^{(3)}_{\text{ind}}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T11**, so the calculus image is of class **D11** (Euler–Lagrange equation and KKT conditions). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Euler–Lagrange equation and KKT conditions of Lemma D11, instantiated at the point $c_3 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D11 needs the feasible set convex — the ℓ_1 ball is — and the objective concave in the relaxed variables, which is the standing assumption on Δ_{8S}. Under those two, KKT is necessary and sufficient; without concavity it is necessary only.

(Q4) Derivation. Relax the integrality of n and form the Lagrangian of the 8S programme. Stationarity in the relaxed variable is the Euler–Lagrange condition; the sign conditions $\lambda, \nu \geq 0$ come from the inequality constraints and complementary slackness records which of them binds. Concavity of the objective on a convex feasible set upgrades necessity to sufficiency.

(Q5) Retraction through the tower. On the lattice the variation becomes a finite difference and the KKT system becomes the optimality conditions of the 8S linear programme over the cross-polytope. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(3)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_3$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The stress when $\varepsilon = 0$ is imposed.

Structure: Setting the total strain to zero and solving gives the constraint reaction — a Lagrange multiplier for the kinematic constraint. The factor $1/(1-\nu)$ is the biaxial modulus, an entry of the constrained compliance.

Reduction: The stress is a constraint multiplier.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T11.

Calculus status: Exact linear; carried to the continuum by Lemma D11, and the input was algebraic, so the differential content is entirely supplied by the transform.

4. Coupled Heat Equation with Deformation

Origin (LA edition).

$$\rho c \dot{T} + T_0 \beta : \dot{\varepsilon} = \nabla \cdot (k \nabla T)$$

Input (8S edition).

$$\left(g^{(5)}_{4j},\ \delta^{(8)}_{4j},\ g^{(3)}_{4j},\ g^{(\mathrm{J})}_{4j}\right) \ \longmapsto\ \mathcal{R}^{8S}_{4j}, \qquad g^{(3)}_{4j} \leq g^{(5)}_{4j}$$

Calculus image.

$$d_g(z_4, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{4j} \leq d_g(z_4, z_j), \qquad g^{(3)}_{4j} \leq g^{(5)}_{4j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is second order — it already carries a Laplacian, a second derivative, or a divergence. The input already carries a second-order operator. The transform therefore does not manufacture the ellipticity of the image; it identifies the input's own second-order part with the principal symbol $-\frac{1}{2}\xi^{\top} G_5^{-1} \xi$ of $\mathcal{L}_\star$, and only the coupling and the gate are new.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_4 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \leq g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{4j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(4)}_{\text{ind}})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_4$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The block system $\left(\begin{smallmatrix} M_{uu} & 0 \\ C_{Tu} & C_{TT} \end{smallmatrix}\right)$ with a non-symmetric coupling.

Structure: The mechanical-to-thermal coupling appears in the damping block while the thermal-to-mechanical coupling sits in the stiffness — the assembled system is unsymmetric unless scaled. Symmetrising it is possible by a diagonal rescaling, which is what monolithic thermoelastic solvers do.

Reduction: Coupling becomes off-diagonal blocks.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input already carried a second-order operator, identified with the principal symbol of $\mathcal{L}_\star$.

5. Thermoelastic Damping (Zener)

Origin (LA edition).

$$Q^{-1} = \frac{E\alpha^2 T_0}{\rho c} \cdot \frac{\omega\tau}{1+\omega^2\tau^2}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi^{(5)}_k = \mu^{(5)}_k \phi^{(5)}_k, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu^{(5)}_1 \le \mu^{(5)}_2 \le \cdots \le \mu^{(5)}_N$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi^{(5)}_k) = \mu^{(5)}_k \phi^{(5)}_k, \qquad \mu^{(5)}_2 = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_5 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(5)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_5$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A Debye peak from a single-pole linear system.

Structure: The coupled system has complex eigenvalues whose imaginary-to-real ratio gives Q; the peak at $\omega\tau = 1$ is where the two blocks exchange energy most efficiently. Damping is an eigenvalue property of the coupled matrix.

Reduction: Damping is a spectral property of the coupled system.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

6. Biot Number

Origin (LA edition).

$$\mathrm{Bi} = \frac{hL}{k}$$

Input (8S edition).

$$\operatorname{rank}(C_5) = \operatorname{rank}(C_4) + \mathbf{1}[\eta^{(6)}_{\text{ind}} > 0], \qquad \dim\ker \mathcal{L}_{8S} = \#\operatorname{comp}(W + \hat{H})$$

Calculus image.

$$\dim\ker \mathcal{L}_\star = \dim H^0(\mathcal{C}) = \#\{\text{components}\}, \qquad \operatorname{ind} \mathcal{L}_\star = 0, \qquad \det \mathrm{Gr}(C_4, e_6) \neq 0 \iff \text{rank increases}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T5**, so the calculus image is of class **D5** (kernel dimension, index, and H^0). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the kernel dimension, index, and H^0 of Lemma D5, instantiated at the point $c_6 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D5 needs $\mathcal{L}_\star$ self-adjoint with compact resolvent, which is Θ-G under **(R1)**–**(R2)**; the identification of ker with locally constant functions needs $G_5 \succ 0$ and $p > 0$, both standing. The index vanishes because the operator is self-adjoint, so coker ≅ ker.

(Q4) Derivation. Set $\mathcal{E}[\phi] = 0$ in Θ-C(1). Since $G_5^{-1} \succ 0$ and $p > 0$, this forces $\nabla\phi = 0$ almost everywhere, hence ϕ locally constant, hence $\ker \mathcal{L}_\star$ spanned by the indicator functions of the connected components — which is precisely H^0. Self-adjointness gives coker ≅ ker and therefore index zero.

(Q5) Retraction through the tower. On the lattice $\dim\ker\mathcal{L}_\star$ becomes $\dim\ker\mathcal{L}_{8S}$, the component count of the coupling graph, and the Gram condition becomes the 8S rank increment. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(6)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_6$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A ratio of boundary to interior conductance.

Structure: $\mathrm{Bi} \ll 1$ means the interior conduction matrix dominates the boundary term, so the temperature field is nearly uniform and the system collapses to one lumped degree of freedom — a rank-one reduction of the conduction matrix.

Reduction: Small Biot number is a rank-one model reduction.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T5.

Calculus status: Exact linear; carried to the continuum by Lemma D5, and the input was algebraic, so the differential content is entirely supplied by the transform.

7. Thermal Shock Resistance Parameter

Origin (LA edition).

$$R = \frac{\sigma_f(1-\nu)}{E\alpha}$$

Input (8S edition).

$$\max_{n_7 \in \mathbb{Z}^5} \; \Delta_{8S} = \mathrm{Score}(\mathcal{M}_8) - \mathrm{Score}(\mathcal{M}_7) \quad \text{s.t.} \quad \|n_7\|_1 \leq M, \;\; \eta^{(7)}_{\text{ind}} \geq \varepsilon_{\text{ind}}$$

Calculus image.

$$\delta\Big[\Delta_{8S} - \lambda(\|n_7\|_1 - M) - \nu(\varepsilon_{\text{ind}} - \eta^{(7)}_{\text{ind}})\Big] = 0, \qquad \lambda, \nu \geq 0, \qquad \lambda(\|n_7\|_1 - M) = \nu(\varepsilon_{\text{ind}} - \eta^{(7)}_{\text{ind}}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T11**, so the calculus image is of class **D11** (Euler–Lagrange equation and KKT conditions). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Euler–Lagrange equation and KKT conditions of Lemma D11, instantiated at the point $c_7 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D11 needs the feasible set convex — the ℓ_1 ball is — and the objective concave in the relaxed variables, which is the standing assumption on Δ_{8S}. Under those two, KKT is necessary and sufficient; without concavity it is necessary only.

(Q4) Derivation. Relax the integrality of n and form the Lagrangian of the 8S programme. Stationarity in the relaxed variable is the Euler–Lagrange condition; the sign conditions $\lambda, \nu \geq 0$ come from the inequality constraints and complementary slackness records which of them binds. Concavity of the objective on a convex feasible set upgrades necessity to sufficiency.

(Q5) Retraction through the tower. On the lattice the variation becomes a finite difference and the KKT system becomes the optimality conditions of the 8S linear programme over the cross-polytope. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(7)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_7$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The temperature drop causing failure under full constraint.

Structure: Inverts the constrained-stress relation of entry 3 at the failure stress. It is a ratio of material constants, each an entry of the stiffness or expansion matrix.

Reduction: The parameter inverts a linear stress relation.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T11.

Calculus status: Exact linear; carried to the continuum by Lemma D11, and the input was algebraic, so the differential content is entirely supplied by the transform.

8. Thermal Buckling Load

Origin (LA edition).

$$\Delta T_{cr} = \frac{\pi^2}{\alpha}\left(\frac{r}{L_e}\right)^2$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(8)} = \mu_k^{(8)}\phi_k^{(8)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(8)} \leq \mu_2^{(8)} \leq \cdots \leq \mu_N^{(8)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(8)}) = \mu_k^{(8)}\phi_k^{(8)}, \qquad \mu_2^{(8)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2\, dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_8 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(8)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_8$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A generalised eigenvalue problem $(K + \Delta T\, K_g)\phi = \mathbf{0}$.

Structure: Buckling is the temperature at which the tangent stiffness first becomes singular — the smallest eigenvalue of the pencil $(K, -K_g)$ with K_g the geometric stiffness from thermal compression. Instability is loss of positive definiteness.

Reduction: Buckling is a generalised eigenproblem.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Structural Engineering.

Poroelasticity and Porous Media

9. Terzaghi Effective Stress

Origin (LA edition).

$$\sigma' = \sigma + pI$$

Input (8S edition).

$$P_5^{(9)} = C_5 C_5^{+} = P_4 + \frac{w_9 w_9^{\top}}{\|w_9\|_2^2}, \qquad w_9 = (I - C_4 C_4^{+}) e_9, \qquad \eta^{(9)}_{\text{ind}} = \frac{\|w_9\|_2^2}{\|e_9\|_2^2 + \varepsilon}$$

Calculus image.

$$\langle e_9 - P_{\mathcal{V}} e_9,\ v\rangle = 0 \quad \forall v \in \mathcal{V}, \qquad \eta^{(9)}_{\text{ind}} = \frac{\det \mathrm{Gr}(C_4, e_9)}{\det \mathrm{Gr}(C_4)(\|e_9\|^2 + \varepsilon)}, \qquad \nabla_{e_9} \|w_9\|^2 = 2\, w_9$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T4**, so the calculus image is of class **D4** (Hilbert projection theorem and Gram volume). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Hilbert projection theorem and Gram volume of Lemma D4, instantiated at the point $c_9 \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D4 needs C_4 of full column rank, so that $\mathrm{Gr}(C_4)$ is invertible and the Schur determinant formula applies. Completeness of $\mathcal{V}$ is needed for the Hilbert projection theorem; a finite-dimensional span is automatically closed, so this costs nothing here.

(Q4) Derivation. Differentiate the least-squares functional: $\frac{d}{ds}\|e - Pe - sv\|^2\big|_{s=0} = -2\langle e - Pe, v\rangle$, and vanishing for all v is the orthogonality condition. For the determinant form, apply the Schur formula to the block Gram matrix, whose Schur complement is $\langle e, (I - P_{\mathcal{V}})e\rangle = \|w\|^2$ by idempotence and self-adjointness of $I - P_{\mathcal{V}}$. The gradient follows by symmetry of $I - P$.

(Q5) Retraction through the tower. On the lattice the inner products become finite sums, the Gram determinants become the matrix Gram determinants, and the ratio becomes the 8S residual $\|w_9\|^2/(\|e_9\|^2 + \varepsilon)$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(9)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_9$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An additive decomposition of total stress.

Structure: Pore pressure contributes only to the spherical part, so the deviatoric stress is unaffected — shear strength depends on σ' alone. This is an orthogonal split into volumetric and deviatoric components.

Reduction: The split is a projection onto the spherical part.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T4.

Calculus status: Exact linear; carried to the continuum by Lemma D4, and the input was algebraic, so the differential content is entirely supplied by the transform.

10. Biot Effective Stress

Origin (LA edition).

$$\sigma' = \sigma + \alpha_B p I, \qquad \alpha_B = 1 - \frac{K}{K_s}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi^{(10)}_k = \mu^{(10)}_k \phi^{(10)}_k, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu^{(10)}_1 \leq \mu^{(10)}_2 \leq \cdots \leq \mu^{(10)}_N$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi^{(10)}_k) = \mu^{(10)}_k \phi^{(10)}_k, \qquad \mu^{(10)}_2 = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{10} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(10)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{10}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Terzaghi with a compressibility-corrected coefficient.

Structure: $\alpha_B \to 1$ for incompressible grains, recovering Terzaghi. The coefficient is a ratio of bulk moduli, hence a ratio of stiffness eigenvalues.

Reduction: The coefficient is a stiffness ratio.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

11. Biot Poroelastic Constitutive Law

Origin (LA edition).

$$\begin{pmatrix} \sigma \\ p \end{pmatrix} = \begin{pmatrix} C & -\alpha_B \mathbf{m} \\ -\alpha_B \mathbf{m}^\top & 1/M \end{pmatrix} \begin{pmatrix} \varepsilon \\ -\zeta \end{pmatrix}$$

Input (8S edition).

$$\text{Admit}_{11}(8S) \iff \eta_{\text{ind}}^{(11)} \geq \varepsilon_{\text{ind}} \ \wedge \ \Delta_{8S} > \varepsilon_{\text{gain}}, \qquad \mu_2(\mathcal{L}_{8S}) = 0 \ \text{ or } \ \tfrac{1}{2}|J_2|\Delta_W + |J_3|\Delta_{\hat{H}} = 1$$

Calculus image.

$$\mathcal{G}_{11}(x, \lambda) = 0, \qquad \partial_x \mathcal{G}_{11} \text{ invertible} \Rightarrow x = x(\lambda) \in C^1; \qquad \text{critical} \iff \partial_x \mathcal{G}_{11} \text{ singular}, \qquad \frac{d\mu_2}{d\lambda}\Big|_{\lambda_c} \neq 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T7**, so the calculus image is of class **D7** (bifurcation and the implicit function theorem). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the bifurcation and the implicit function theorem of Lemma D7, instantiated at the point $c_{11} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D7 needs $\mathcal{G}$ to be C^1 in both arguments for the implicit function theorem, and transversality $d\mu_2/d\lambda \neq 0$ at the crossing for the bifurcation to be simple. Where transversality fails the crossing is degenerate and the branch structure is not a fold.

(Q4) Derivation. Write the entry's critical condition as $\mathcal{G}(x, \lambda) = 0$. The implicit function theorem gives a unique C^1 branch $x(\lambda)$ exactly while $\partial_x \mathcal{G}$ is invertible; the threshold is where invertibility fails. For a spectral threshold that is $\mu_2 = 0$ by Θ-G, and for a gate threshold it is the loss of strict convexity of $\mathcal{F}$ by Θ-H(2). Differentiating the eigenvalue along the branch gives the transversality condition.

(Q5) Retraction through the tower. On the lattice the implicit-function branch becomes the 8S parameter branch and the singularity condition becomes whichever of the three 8S thresholds this entry matched. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(11)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{11}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A symmetric block matrix; **m** the Voigt identity vector.

Structure: Symmetry of the off-diagonal blocks is the poroelastic reciprocity — the same coefficient α_B appears in both couplings, which is an Onsager relation. Positive definiteness of the whole block matrix is thermodynamic admissibility.

Reduction: The constitutive law is a symmetric block matrix.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T7.

Calculus status: Exact linear; carried to the continuum by Lemma D7, and the input was algebraic, so the differential content is entirely supplied by the transform.

12. Fluid Content Storage Equation

Origin (LA edition).

$$\frac{\partial \zeta}{\partial t} = \nabla \cdot \left(\frac{k}{\mu} \nabla p \right)$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(12)} = \mu_k^{(12)} \phi_k^{(12)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(12)} \le \mu_2^{(12)} \le \cdots \le \mu_N^{(12)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(12)}) = \mu_k^{(12)} \phi_k^{(12)}, \qquad \mu_2^{(12)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is second order — it already carries a Laplacian, a second derivative, or a divergence. The input already carries a second-order operator. The transform therefore does not manufacture the ellipticity of the image; it identifies the input's own second-order part with the principal symbol $-\frac{1}{2}\xi^\top G_5^{-1} \xi$ of $\mathcal{L}_\star$, and only the coupling and the gate are new.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{12} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(12)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{12}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A diffusion equation for pore pressure coupled to volumetric strain.

Structure: Combining with the constitutive law gives $\dot{p}$ driven by both a Laplacian and $\dot{\varepsilon}_v$ — the coupled system $\left(\begin{smallmatrix} K & -Q \\ Q^\top & S \end{smallmatrix}\right)$ is symmetric indefinite, exactly the KKT structure of a saddle-point problem.

Reduction: The coupled system is a symmetric saddle-point matrix.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input already carried a second-order operator, identified with the principal symbol of $\mathcal{L}_\star$.

13. Darcy's Law

Origin (LA edition).

$$\mathbf{q} = -\frac{k}{\mu}\nabla p$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(13)} = \mu_k^{(13)}\phi_k^{(13)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(13)} \le \mu_2^{(13)} \le \cdots \le \mu_N^{(13)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(13)}) = \mu_k^{(13)}\phi_k^{(13)}, \qquad \mu_2^{(13)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{13} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(13)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{13}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A permeability tensor mapping pressure gradient to flux.

Structure: k is symmetric positive definite for a physically admissible medium — symmetry is an Onsager relation, positive definiteness is the second law. Anisotropic media have distinct eigenvalues, giving preferential flow directions.

Reduction: Flux is a matrix times a gradient.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input already carried a first-order operator, which the transform leaves in place.

Also in: Hydrology, Chemical Engineering.

14. Terzaghi Consolidation Equation

Origin (LA edition).

$$\frac{\partial u}{\partial t} = c_v \frac{\partial^2 u}{\partial z^2}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(14)} = \mu_k^{(14)}\phi_k^{(14)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(14)} \le \mu_2^{(14)} \le \cdots \le \mu_N^{(14)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(14)}) = \mu_k^{(14)}\phi_k^{(14)}, \qquad \mu_2^{(14)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is second order — it already carries a Laplacian, a second derivative, or a divergence. The input already carries a second-order operator. The transform therefore does not manufacture the ellipticity of the image; it identifies the input's own second-order part with the principal symbol $-\frac{1}{2}\xi^\top G_5^{-1}\xi$ of $\mathcal{L}_\star$, and only the coupling and the gate are new.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{14} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(14)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{14}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A diffusion equation for excess pore pressure.

Structure: Identical in form to the heat equation, so the discretised operator is the symmetric negative semidefinite Laplacian and the solution is $e^{-c_v L t}\mathbf{u}_0$. Consolidation time scales as L^2/c_v from the smallest nonzero eigenvalue.

Reduction: Consolidation is a matrix exponential.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input already carried a second-order operator, identified with the principal symbol of $\mathcal{L}_\star$.

15. Degree of Consolidation

Origin (LA edition).

$$U = 1 - \sum_m \frac{2}{M^2} e^{-M^2 T_v}, \qquad M = \frac{(2m+1)\pi}{2}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(15)} = \mu_k^{(15)}\phi_k^{(15)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(15)} \le \mu_2^{(15)} \le \cdots \le \mu_N^{(15)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(15)}) = \mu_k^{(15)}\phi_k^{(15)}, \qquad \mu_2^{(15)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{15} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(15)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{15}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A modal expansion in the Laplacian eigenbasis.

Structure: Each term is one eigenmode decaying at its own rate; the series is the spectral decomposition of e^{-Lt} applied to the initial condition. Higher modes decay fastest, so the response is soon dominated by the first.

Reduction: The series is a spectral expansion.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

16. Skempton Coefficient

Origin (LA edition).

$$B = \frac{\Delta p}{\Delta \sigma_{\mathrm{oct}}}$$

Input (8S edition).

$$y_{16}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{16})\Big] z(u), \qquad z_{16}(u) = \begin{bmatrix} c_{16} \\ r_{16} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{16})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{16}\, \psi_{16}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{16} = -\mathcal{L}_\star \psi_{16}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{16} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{16})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(16)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{16}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An off-diagonal ratio in the undrained compliance matrix.

Structure: The pressure response to applied stress at constant fluid content — obtained by taking a Schur complement of the poroelastic block matrix with respect to the fluid degree of freedom. Undrained behaviour is a condensation.

Reduction: The coefficient is a Schur complement entry.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

17. Poroelastic Wave Speeds (Biot)

Origin (LA edition).

fast P, slow P, and S waves

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(17)} = \mu_k^{(17)}\phi_k^{(17)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(17)} \le \mu_2^{(17)} \le \cdots \le \mu_N^{(17)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(17)}) = \mu_k^{(17)}\phi_k^{(17)}, \qquad \mu_2^{(17)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{17} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(17)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{17}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Eigenvalues of a 2×2 (or 3×3) coupled Christoffel matrix.

Structure: Coupling the solid and fluid momentum equations gives a generalised eigenproblem whose three roots are the two compressional speeds and one shear speed. The slow P wave is diffusive at low frequency — its eigenvalue has a large imaginary part.

Reduction: Wave speeds are eigenvalues of a coupled matrix.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

18. Kozeny–Carman Permeability

Origin (LA edition).

$$k = \frac{\phi^3}{c\, S^2 (1-\phi)^2}$$

Input (8S edition).

$$\neg\,\mathrm{Admit}_{18}(8S) \iff \eta^{(18)}_{\mathrm{ind}} < \varepsilon_{\mathrm{ind}} \ \vee\ \Delta_{8S} \le \varepsilon_{\mathrm{gain}}, \qquad e_{18} \in \mathcal{R}(C_4)$$

Calculus image.

$$\delta\Delta_{8S}[e_{18}] = 0 \ \text{ for every admissible variation } \iff (I - P_{\mathcal{V}})e_{18} = 0 \iff \det \mathrm{Gr}(C_4, e_{18}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T14**, so the calculus image is of class **D14** (vanishing first variation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the vanishing first variation of Lemma D14, instantiated at the point $c_{18} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D14 needs nothing beyond Θ-A(3), the identity $\nabla_e \|w\|^2 = 2(I - P_{\mathcal{V}})e$. In particular no regularity, no bandwidth limit, and no property of the kernel is used — appropriately, since the claim is that there is no differential content to regularise.

(Q4) Derivation. Compute the first variation of the score in the direction of the entry's own content. By Θ-A(3) it is $2(I - P_{\mathcal{V}})e$, which vanishes identically precisely when $e \in \mathcal{V}$. Equivalently, by Θ-A(4), the Gram determinant vanishes and the parallelepiped spanned by C_4 and e has zero five-dimensional volume. There is therefore nothing to differentiate, and the calculus image is the statement that the variation is identically zero.

(Q5) Retraction through the tower. On the lattice the vanishing variation becomes $\eta^{(18)}_{\mathrm{ind}} = 0$, the 8S rejection certificate. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. The entry is **not** admitted, and the calculus image says why in a third vocabulary: the first variation of the score in the direction of this entry's content vanishes identically, so the Gâteaux derivative carries no information and the Gram volume $\det \mathrm{Gr}(C_4, e_{18})$ is zero. Differentiating an obstruction returns zero, and that is the correct output.

Objects: A scalar constitutive correlation.

Structure: An empirical relation between geometry and the permeability tensor's magnitude. Nonlinear in porosity; only the resulting k enters linearly through Darcy's law.

Reduction: Not linear in the porosity.

Status: Structural analogue.

8S status: Rejection certificate (Lemma T14) — the obstruction is preserved, not removed.

Calculus status: Vanishing first variation (Lemma D14) — the obstruction survives its third translation intact.

19. Richards Equation

Origin (LA edition).

$$\frac{\partial \theta}{\partial t} = \nabla \cdot (K(\theta) \nabla h)$$

Input (8S edition).

$$W_{19j} = A_{19j} \exp\left[-\frac{(c_{19} - c_j)^\top G_5 (c_{19} - c_j)}{2\sigma_c^2} - \frac{d_{\mathrm{J}}(19, j)^2}{2\sigma_{\mathrm{J}}^2} \right], \qquad W \succeq 0$$

Calculus image.

$$\partial_t k_t = \nabla \cdot (G_5^{-1} \nabla k_t), \quad k_0 = \delta_0; \qquad W_\sigma\big|_{\text{node } 19} = e^{t\mathcal{A}_G}, \quad t = \tfrac{\sigma_c^2}{2}; \qquad \widehat{k_t}(\xi) = e^{-t\,\xi^\top G_5^{-1} \xi}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T8**, so the calculus image is of class **D8** (heat semigroup and its symbol). The originating equation is second order — it already carries a Laplacian, a second derivative, or a divergence. The input already carries a second-order operator. The transform therefore does not manufacture the ellipticity of the image; it identifies the input's own second-order part with the principal symbol $-\frac{1}{2}\xi^\top G_5^{-1}\xi$ of $\mathcal{L}_\star$, and only the coupling and the gate are new.

(Q2) Image. The displayed relation is the heat semigroup and its symbol of Lemma D8, instantiated at the point $c_{19} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D8 needs $G_5 \succ 0$ for the Gaussian to be integrable and for $G_5^{1/2}$ to exist. No hypothesis on the tag factor is required for the heat-kernel identification, since that factor is a smooth positive multiplier; where it is retained, it is absorbed into the sampling density p.

(Q4) Derivation. Substitute $y = G_5^{1/2} x$, under which $\nabla \cdot (G_5^{-1} \nabla \cdot)$ becomes the ordinary Laplacian and the coupling kernel becomes the standard Gaussian. Differentiating $t^{-5/2} e^{-\|y\|^2/4t}$ in t and in y verifies the heat equation directly. Matching exponents $\|x\|_{G_5}^2 / 2\sigma_c^2 = \|x\|_{G_5}^2 / 4t$ identifies $t = \sigma_c^2/2$. The symbol follows from the Gaussian Fourier integral $\int e^{-x^\top A x} e^{-i\xi \cdot x} dx = \pi^{5/2} \det(A)^{-1/2} e^{-\xi^\top A^{-1} \xi / 4}$.

(Q5) Retraction through the tower. Sampling the heat kernel at lattice separations returns W_{ij} exactly, since the 8S kernel *is* k_t evaluated at $c_i - c_j$ with $t = \sigma_c^2/2$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(19)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{19}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A diffusion equation with solution-dependent conductivity.

Structure: Genuinely nonlinear, since K depends on θ. Newton linearisation gives a sequence of symmetric positive definite systems, each with the current $K(\theta)$ frozen — the standard solution strategy.

Reduction: Newton linearisation gives linear systems.

Status: Linearised.

8S status: Linearised; the 8S image inherits the same approximation status (Theorem C, Lemma T8).

Calculus status: Linearised; carried to the continuum by Lemma D8, and the input already carried a second-order operator, identified with the principal symbol of $\mathcal{L}_\star$.

Also in: Hydrology.

Fluid–Structure Interaction

20. Kinematic Interface Condition

Origin (LA edition).

$$\mathbf{v}_f = \dot{\mathbf{u}}_s \text{ on } \Gamma$$

Input (8S edition).

$$\operatorname{rank}(C_5) = \operatorname{rank}(C_4) + \mathbf{1}[\eta_{\text{ind}}^{(20)} > 0], \qquad \dim\ker \mathcal{L}_{8S} = \#\operatorname{comp}(W + \hat{H})$$

Calculus image.

$$\dim\ker \mathcal{L}_\star = \dim H^0(\mathcal{C}) = \#\{\text{components}\}, \qquad \operatorname{ind} \mathcal{L}_\star = 0, \qquad \det \operatorname{Gr}(C_4, e_{20}) \neq 0 \iff \text{rank increases}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T5**, so the calculus image is of class **D5** (kernel dimension, index, and H^0). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the kernel dimension, index, and H^0 of Lemma D5, instantiated at the point $c_{20} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D5 needs $\mathcal{L}_\star$ self-adjoint with compact resolvent, which is Θ-G under **(R1)**–**(R2)**; the identification of ker with locally constant functions needs $G_5 \succ 0$ and $p > 0$, both standing. The index vanishes because the operator is self-adjoint, so coker ≅ ker.

(Q4) Derivation. Set $\mathcal{E}[\phi] = 0$ in Θ-C(1). Since $G_5^{-1} \succ 0$ and $p > 0$, this forces $\nabla\phi = 0$ almost everywhere, hence ϕ locally constant, hence $\ker \mathcal{L}_\star$ spanned by the indicator functions of the connected components — which is precisely H^0. Self-adjointness gives coker ≅ ker and therefore index zero.

(Q5) Retraction through the tower. On the lattice $\dim\ker \mathcal{L}_\star$ becomes $\dim\ker \mathcal{L}_{8S}$, the component count of the coupling graph, and the Gram condition becomes the 8S rank increment. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(20)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{20}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A linear constraint coupling two field variables.

Structure: Enforced by Lagrange multipliers (mortar methods) giving a saddle-point block system, or by direct elimination. The constraint matrix is the interface trace operator, and its rank is the number of interface degrees of freedom.

Reduction: The condition is a linear constraint.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T5.

Calculus status: Exact linear; carried to the continuum by Lemma D5, and the input already carried a first-order operator, which the transform leaves in place.

21. Dynamic Interface Condition

Origin (LA edition).

$$\sigma_f \mathbf{n} = \sigma_s \mathbf{n} \text{ on } \Gamma$$

Input (8S edition).

$$\max_{n_{21} \in \mathbb{Z}^5} \Delta_{8S} = \mathrm{Score}(\mathcal{M}_8) - \mathrm{Score}(\mathcal{M}_7) \quad \text{s.t.} \quad \|n_{21}\|_1 \le M, \;\; \eta_{\mathrm{ind}}^{(21)} \ge \varepsilon_{\mathrm{ind}}$$

Calculus image.

$$\delta\Big[\Delta_{8S} - \lambda(\|n_{21}\|_1 - M) - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(21)})\Big] = 0, \qquad \lambda, \nu \ge 0, \qquad \lambda(\|n_{21}\|_1 - M) = \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(21)}) = 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T11**, so the calculus image is of class **D11** (Euler–Lagrange equation and KKT conditions). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Euler–Lagrange equation and KKT conditions of Lemma D11, instantiated at the point $c_{21} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D11 needs the feasible set convex — the ℓ_1 ball is — and the objective concave in the relaxed variables, which is the standing assumption on Δ_{8S}. Under those two, KKT is necessary and sufficient; without concavity it is necessary only.

(Q4) Derivation. Relax the integrality of n and form the Lagrangian of the 8S programme. Stationarity in the relaxed variable is the Euler–Lagrange condition; the sign conditions $\lambda, \nu \ge 0$ come from the inequality constraints and complementary slackness records which of them binds. Concavity of the objective on a convex feasible set upgrades necessity to sufficiency.

(Q5) Retraction through the tower. On the lattice the variation becomes a finite difference and the KKT system becomes the optimality conditions of the 8S linear programme over the cross-polytope. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(21)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{21}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Traction continuity — the dual of the kinematic condition.

Structure: The two interface conditions are adjoint: the kinematic constraint matrix and the traction transfer matrix are transposes. This transposition is what makes monolithic FSI systems symmetric when the physics permits.

Reduction: The condition is the transpose of the kinematic constraint.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T11.

Calculus status: Exact linear; carried to the continuum by Lemma D11, and the input was algebraic, so the differential content is entirely supplied by the transform.

22. Added Mass Coefficient

Origin (LA edition).

$$M_{\mathrm{eff}} = M_s + M_a$$

Input (8S edition).

$$W_{22j} = A_{22j} \exp\left[-\frac{(c_{22}-c_j)^\top G_5 (c_{22}-c_j)}{2\sigma_c^2} - \frac{d_{\mathrm{J}}(22,j)^2}{2\sigma_{\mathrm{J}}^2} \right], \qquad W \succeq 0$$

Calculus image.

$$\partial_t k_t = \nabla\cdot(G_5^{-1}\nabla k_t), \quad k_0 = \delta_0; \qquad W_\sigma\big|_{\text{node } 22} = e^{t\mathcal{A}_G}, \quad t = \tfrac{\sigma_c^2}{2}; \qquad \widehat{k_t}(\xi) = e^{-t\,\xi^\top G_5^{-1}\xi}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T8**, so the calculus image is of class **D8** (heat semigroup and its symbol). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the heat semigroup and its symbol of Lemma D8, instantiated at the point $c_{22} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D8 needs $G_5 \succ 0$ for the Gaussian to be integrable and for $G_5^{1/2}$ to exist. No hypothesis on the tag factor is required for the heat-kernel identification, since that factor is a smooth positive multiplier; where it is retained, it is absorbed into the sampling density p.

(Q4) Derivation. Substitute $y = G_5^{1/2}x$, under which $\nabla\cdot(G_5^{-1}\nabla\cdot)$ becomes the ordinary Laplacian and the coupling kernel becomes the standard Gaussian. Differentiating $t^{-5/2}e^{-\|y\|^2/4t}$ in t and in y verifies the heat equation directly. Matching exponents $\|x\|^2_{G_5}/2\sigma_c^2 = \|x\|^2_{G_5}/4t$ identifies $t = \sigma_c^2/2$. The symbol follows from the Gaussian Fourier integral $\int e^{-x^\top Ax}e^{-i\xi\cdot x}dx = \pi^{5/2}\det(A)^{-1/2}e^{-\xi^\top A^{-1}\xi/4}$.

(Q5) Retraction through the tower. Sampling the heat kernel at lattice separations returns W_{ij} exactly, since the 8S kernel *is* k_t evaluated at $c_i - c_j$ with $t = \sigma_c^2/2$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\,I_3\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(22)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{22}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An additive modification of the mass matrix.

Structure: The fluid's inertial reaction appears as an extra positive definite mass matrix, obtained by condensing out the fluid pressure — a Schur complement. For dense fluids M_a can exceed M_s, which destabilises partitioned solvers.

Reduction: Added mass is a Schur complement of the fluid block.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T8.

Calculus status: Exact linear; carried to the continuum by Lemma D8, and the input was algebraic, so the differential content is entirely supplied by the transform.

23. Reduced Velocity

Origin (LA edition).

$$U_r = \frac{U}{f_n D}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(23)} = \mu_k^{(23)}\phi_k^{(23)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(23)} \leq \mu_2^{(23)} \leq \cdots \leq \mu_N^{(23)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(23)}) = \mu_k^{(23)}\phi_k^{(23)}, \qquad \mu_2^{(23)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{23} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(23)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{23}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A ratio comparing flow and structural timescales.

Structure: f_n comes from the structural eigenproblem $(K - \omega^2 M)\phi = \mathbf{0}$, so the parameter compares a flow velocity to a spectral quantity. Lock-in occurs when the shedding frequency approaches an eigenvalue.

Reduction: The frequency is an eigenvalue.

Status: Exact linear in the frequency.

8S status: Exact linear in the frequency; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear in the frequency; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

24. Strouhal Number

Origin (LA edition).

$$\mathrm{St} = \frac{f_s D}{U} \approx 0.2$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(24)} = \mu_k^{(24)}\phi_k^{(24)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(24)} \leq \mu_2^{(24)} \leq \cdots \leq \mu_N^{(24)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(24)}) = \mu_k^{(24)}\phi_k^{(24)}, \qquad \mu_2^{(24)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{24} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(24)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{24}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The shedding frequency of the wake instability.

Structure: Arises as the imaginary part of the least stable eigenvalue of the linearised Navier–Stokes operator about the steady wake — global stability analysis. The remarkable constancy near 0.2 is the near-invariance of that eigenvalue over a wide Reynolds range.

Reduction: The frequency is an eigenvalue of the linearised operator.

Status: Linearised (global stability).

8S status: Linearised (global stability); the 8S image inherits the same approximation status (Theorem C, Lemma T2).

Calculus status: Linearised (global stability); carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Fluid Dynamics.

25. Flutter Speed (Binary Model)

Origin (LA edition).

$$\det(-\omega^2 M + i\omega C + K + qA) = 0$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(25)} = \mu_k^{(25)}\phi_k^{(25)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(25)} \leq \mu_2^{(25)} \leq \cdots \leq \mu_N^{(25)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(25)}) = \mu_k^{(25)}\phi_k^{(25)}, \qquad \mu_2^{(25)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{25} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(25)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{25}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A quadratic eigenvalue problem with a speed-dependent aerodynamic matrix.

Structure: Flutter is the dynamic pressure q at which an eigenvalue crosses the imaginary axis — a Hopf bifurcation in the coupled system. The aerodynamic matrix A is non-symmetric, which is precisely what permits energy extraction from the flow.

Reduction: Flutter is an eigenvalue crossing.

Status: Exact linear (eigenvalue problem).

8S status: Exact linear (eigenvalue problem); preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear (eigenvalue problem); carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Aerospace Engineering.

26. Divergence Speed

Origin (LA edition).

$$\det(K + qA_s) = 0$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(26)} = \mu_k^{(26)}\phi_k^{(26)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(26)} \leq \mu_2^{(26)} \leq \cdots \leq \mu_N^{(26)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(26)}) = \mu_k^{(26)}\phi_k^{(26)}, \qquad \mu_2^{(26)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{26} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(26)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{26}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A static instability — singularity of the coupled stiffness.

Structure: Divergence is loss of invertibility of $K + qA_s$, the static analogue of buckling. It is a generalised eigenproblem for the critical dynamic pressure, with the null vector giving the divergence mode shape.

Reduction: Divergence is a generalised eigenproblem.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

27. Theodorsen Function

Origin (LA edition).

$$C(k) = F(k) + iG(k)$$

Input (8S edition).

$$y_{27}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{27})\Big] z(u), \qquad z_{27}(u) = \begin{bmatrix} c_{27} \\ r_{27} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma\psi_{27})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c,c')\, \gamma(c')\, \mathsf{A}_{27}\, \psi_{27}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{27} = -\mathcal{L}_\star \psi_{27}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{27} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C}\times\mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j(\cdot)_j$ by $\int_{\mathcal{C}}(\cdot)\,p\,dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\,uu^\top du = \sigma^2\kappa\,G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{27})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2,K_3,J_2,J_3\to 0$ and $\Pi\to[\,I_3\;0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(27)}_{\text{ind}})]$ vanishes with $\nu\ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{27}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A frequency-domain transfer function for unsteady lift.

Structure: A complex multiplier on the quasi-steady lift, capturing the wake's memory. Being causal, its real and imaginary parts satisfy Kramers–Kronig — related by the Hilbert transform matrix.

Reduction: The function is a frequency-domain multiplier.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

28. Womersley Number

Origin (LA edition).

$$\mathrm{Wo} = R\sqrt{\frac{\omega\rho}{\mu}}$$

Input (8S edition).

$$E_{28}^{8S}(z) = z^\top \mathcal{G}_8 z\Big|_{\text{node }28}, \qquad \big\|z_{28}(u) - z_j(u)\big\|_{\mathcal{G}_8} = \delta^{(8)}_{28j} = \sqrt{(c_{28}-c_j)^\top G_5 (c_{28}-c_j) + (r_{28}-r_j)^2}$$

Calculus image.

$$\mathcal{E}_{28}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta\mathcal{E}_{28} = 0 \iff \mathcal{L}_\star\phi = 0, \qquad ds^2_{28} = dc^\top(G_5 + \nabla r\,\nabla r^\top)dc + r_{28}^2\, d\Omega_3^2$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{28} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_\star \phi, \phi \rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2}\int \|\nabla\phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c,u) = (c, r(c)u)$, use $u{\cdot}du = \frac{1}{2}d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r{\cdot}dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(28)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{28}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A ratio of unsteady inertia to viscous diffusion.

Structure: Large Wo gives a plug-like profile with a thin Stokes layer; small Wo gives quasi-Poiseuille flow. The velocity profile is a Bessel-function eigenmode, so the parameter selects which eigenmode dominates.

Reduction: The parameter selects a modal regime.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T3.

Calculus status: Exact linear; carried to the continuum by Lemma D3, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Biophysics.

29. Moens–Korteweg Pulse Wave Velocity

Origin (LA edition).

$$c = \sqrt{\frac{Eh}{2\rho R}}$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(29)} = \mu_k^{(29)} \phi_k^{(29)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(29)} \leq \mu_2^{(29)} \leq \cdots \leq \mu_N^{(29)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1} \nabla \phi_k^{(29)}) = \mu_k^{(29)} \phi_k^{(29)}, \qquad \mu_2^{(29)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{29} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(29)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{29}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A wave speed from a coupled fluid–wall system.

Structure: Emerges as the eigenvalue of a 2×2 hyperbolic system coupling flow rate and area, with wall compliance supplying the restoring term. Stiffer walls raise the speed, which is the basis of arterial stiffness measurement.

Reduction: The speed is an eigenvalue of a coupled hyperbolic system.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Biophysics.

30. Water Hammer (Joukowsky) Pressure Rise

Origin (LA edition).

$$\Delta p = \rho c \, \Delta v$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(30)} = \mu_k^{(30)} \phi_k^{(30)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(30)} \leq \mu_2^{(30)} \leq \cdots \leq \mu_N^{(30)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(30)}) = \mu_k^{(30)} \phi_k^{(30)}, \qquad \mu_2^{(30)} = \min_{\substack{\phi \neq 0 \\ \int \phi \, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}} \, p^2 \, dc}{\int_{\mathcal{C}} \phi^2 \, p^2 \, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{30} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(30)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{30}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A Riemann invariant relation across a characteristic.

Structure: ρc is the acoustic impedance — the off-diagonal entry linking pressure and velocity in the 2×2 characteristic system. Pressure and velocity jumps are coupled by exactly this impedance across each wave.

Reduction: The relation is a characteristic (eigenvector) condition.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

Electro- and Magneto-Mechanical Coupling

31. Piezoelectric Constitutive Equations

Origin (LA edition).

$$\begin{pmatrix} \sigma \\ \mathbf{D} \end{pmatrix} = \begin{pmatrix} C^E & -e^\top \\ e & \epsilon^S \end{pmatrix} \begin{pmatrix} \varepsilon \\ \mathbf{E} \end{pmatrix}$$

Input (8S edition).

$$E^{8S}_{31}(z) = z^\top \mathcal{G}_8 z \Big|_{\text{node } 31}, \qquad \big\| z_{31}(u) - z_j(u) \big\|_{\mathcal{G}_8} = \delta^{(8)}_{31j} = \sqrt{(c_{31} - c_j)^\top G_5 (c_{31} - c_j) + (r_{31} - r_j)^2}$$

Calculus image.

$$\mathcal{E}_{31}[\phi] = \tfrac{1}{2} \int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}} \, p^2 \, dc, \qquad \delta \mathcal{E}_{31} = 0 \iff \mathcal{L}_\star \phi = 0, \qquad ds^2_{31} = dc^\top (G_5 + \nabla r \, \nabla r^\top) dc + r^2_{31} \, d\Omega^2_3$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{31} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_\star \phi, \phi \rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2} \int \|\nabla \phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c, u) = (c, r(c)u)$, use $u \cdot du = \frac{1}{2} d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r \cdot dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(31)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{31}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A block matrix with the piezoelectric tensor e off-diagonal.

Structure: The off-diagonal blocks are transposes of each other up to sign — a reciprocity relation guaranteeing that the coupled system derives from a single energy function. Direct and converse piezoelectric effects are the two off-diagonal blocks.

Reduction: Coupling is an off-diagonal block pair.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T3.

Calculus status: Exact linear; carried to the continuum by Lemma D3, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Electrical Engineering.

32. Electromechanical Coupling Factor

Origin (LA edition).

$$k^2 = \frac{e^2}{C^E \epsilon^S}$$

Input (8S edition).

$$P_5^{(32)} = C_5 C_5^+ = P_4 + \frac{w_{32} w_{32}^\top}{\|w_{32}\|_2^2}, \qquad w_{32} = (I - C_4 C_4^+) e_{32}, \qquad \eta^{(32)}_{\text{ind}} = \frac{\|w_{32}\|_2^2}{\|e_{32}\|_2^2 + \varepsilon}$$

Calculus image.

$$\langle e_{32} - P_{\mathcal{V}} e_{32},\ v \rangle = 0 \quad \forall v \in \mathcal{V}, \qquad \eta^{(32)}_{\text{ind}} = \frac{\det \text{Gr}(C_4, e_{32})}{\det \text{Gr}(C_4)(\|e_{32}\|^2 + \varepsilon)}, \qquad \nabla_{e_{32}} \|w_{32}\|^2 = 2\, w_{32}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T4**, so the calculus image is of class **D4** (Hilbert projection theorem and Gram volume). The originating equation is algebraic (order 0). The input is algebraic,

so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Hilbert projection theorem and Gram volume of Lemma D4, instantiated at the point $c_{32} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D4 needs C_4 of full column rank, so that $\mathrm{Gr}(C_4)$ is invertible and the Schur determinant formula applies. Completeness of $\mathcal{V}$ is needed for the Hilbert projection theorem; a finite-dimensional span is automatically closed, so this costs nothing here.

(Q4) Derivation. Differentiate the least-squares functional: $\frac{d}{ds}\|e - Pe - sv\|^2\big|_{s=0} = -2\langle e - Pe, v\rangle$, and vanishing for all v is the orthogonality condition. For the determinant form, apply the Schur formula to the block Gram matrix, whose Schur complement is $\langle e, (I - P_{\mathcal{V}})e\rangle = \|w\|^2$ by idempotence and self-adjointness of $I - P_{\mathcal{V}}$. The gradient follows by symmetry of $I - P$.

(Q5) Retraction through the tower. On the lattice the inner products become finite sums, the Gram determinants become the matrix Gram determinants, and the ratio becomes the 8S residual $\|w_{32}\|^2/(\|e_{32}\|^2 + \varepsilon)$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta^{(32)}_{\mathrm{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{32}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A normalised measure of off-diagonal dominance.

Structure: Exactly the ratio of the off-diagonal product to the diagonal product in the 2×2 block matrix — the same quantity that controls the eigenvalue splitting. Positive definiteness requires $k^2 < 1$, so the bound is a definiteness condition.

Reduction: The factor measures block off-diagonal dominance.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T4.

Calculus status: Exact linear; carried to the continuum by Lemma D4, and the input was algebraic, so the differential content is entirely supplied by the transform.

33. Electrostrictive Strain

Origin (LA edition).

$$\varepsilon = QE^2$$

Input (8S edition).

$$E^{8S}_{33}(z) = z^\top \mathcal{G}_8 z\Big|_{\text{node 33}}, \qquad \|z_{33}(u) - z_j(u)\|_{\mathcal{G}_8} = \delta^{(8)}_{33j} = \sqrt{(c_{33} - c_j)^\top G_5 (c_{33} - c_j) + (r_{33} - r_j)^2}$$

Calculus image.

$$\mathcal{E}_{33}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta\mathcal{E}_{33} = 0 \iff \mathcal{L}_\star \phi = 0, \qquad ds^2_{33} = dc^\top (G_5 + \nabla r\, \nabla r^\top) dc + r^2_{33}\, d\Omega^2_3$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{33} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_\star \phi, \phi \rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2}\int \|\nabla\phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c,u) = (c, r(c)u)$, use $u \cdot du = \frac{1}{2} d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r \cdot dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(33)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{33}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Quadratic in the field, hence not linear.

Structure: Present in all dielectrics, unlike piezoelectricity which requires broken centrosymmetry. Being quadratic, it produces no linear coupling block — linearising about a bias field E_0 gives an effective piezoelectric coefficient $2QE_0$.

Reduction: Linearisation about a bias field gives a linear coupling.

Status: Linearised.

8S status: Linearised; the 8S image inherits the same approximation status (Theorem C, Lemma T3).

Calculus status: Linearised; carried to the continuum by Lemma D3, and the input was algebraic, so the differential content is entirely supplied by the transform.

34. Maxwell Stress Tensor

Origin (LA edition).

$$T_{ij} = \epsilon \left(E_i E_j - \tfrac{1}{2} \delta_{ij} E^2 \right)$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(34)} = \mu_k^{(34)} \phi_k^{(34)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(34)} \leq \mu_2^{(34)} \leq \cdots \leq \mu_N^{(34)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(34)}) = \mu_k^{(34)} \phi_k^{(34)}, \qquad \mu_2^{(34)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{34} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(34)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{34}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A symmetric traceless-plus-isotropic matrix quadratic in the field.

Structure: Quadratic in $\mathbf{E}$, so the mechanical response is nonlinear in the field. The tensor is symmetric with eigenvalues $(\frac{1}{2}\epsilon E^2, -\frac{1}{2}\epsilon E^2, -\frac{1}{2}\epsilon E^2)$ — tension along the field lines and pressure across them.

Reduction: Quadratic in the field; the tensor itself is a matrix.

Status: Exact linear as a tensor, quadratic in the field.

8S status: Exact linear as a tensor, quadratic in the field; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear as a tensor, quadratic in the field; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Electromagnetism.

35. Electrostatic Pull-In Voltage

Origin (LA edition).

$$V_{PI} = \sqrt{\frac{8kg_0^3}{27\epsilon A}}$$

Input (8S edition).

$$\text{Admit}_{35}(8S) \iff \eta_{\text{ind}}^{(35)} \geq \varepsilon_{\text{ind}} \;\wedge\; \Delta_{8S} > \varepsilon_{\text{gain}}, \qquad \mu_2(\mathcal{L}_{8S}) = 0 \;\text{ or }\; \tfrac{1}{2}|J_2|\Delta_W + |J_3|\Delta_{\hat{H}} = 1$$

Calculus image.

$$\mathcal{G}_{35}(x, \lambda) = 0, \qquad \partial_x \mathcal{G}_{35} \text{ invertible} \Rightarrow x = x(\lambda) \in C^1; \qquad \text{critical} \iff \partial_x \mathcal{G}_{35} \text{ singular}, \qquad \frac{d\mu_2}{d\lambda}\Big|_{\lambda_c} \neq 0$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T7**, so the calculus image is of class **D7** (bifurcation and the implicit function theorem). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the bifurcation and the implicit function theorem of Lemma D7, instantiated at the point $c_{35} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D7 needs $\mathcal{G}$ to be C^1 in both arguments for the implicit function theorem, and transversality $d\mu_2/d\lambda \neq 0$ at the crossing for the bifurcation to be simple. Where transversality fails the crossing is degenerate and the branch structure is not a fold.

(Q4) Derivation. Write the entry's critical condition as $\mathcal{G}(x, \lambda) = 0$. The implicit function theorem gives a unique C^1 branch $x(\lambda)$ exactly while $\partial_x \mathcal{G}$ is invertible; the threshold is where invertibility fails. For a spectral threshold that is $\mu_2 = 0$ by Θ-G, and for a gate threshold it is the loss of strict convexity of $\mathcal{F}$ by Θ-H(2). Differentiating the eigenvalue along the branch gives the transversality condition.

(Q5) Retraction through the tower. On the lattice the implicit-function branch becomes the 8S parameter branch and the singularity condition becomes whichever of the three 8S thresholds this entry matched. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(35)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{35}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A saddle-node bifurcation in the coupled system.

Structure: The effective stiffness $k - \partial F_e/\partial g$ vanishes at pull-in — the tangent stiffness matrix becomes singular. Instability at one third of the gap is where the linearised system first loses positive definiteness.

Reduction: Pull-in is loss of definiteness of the tangent stiffness.

Status: Exact linear at the stability condition.

8S status: Exact linear at the stability condition; preserved under the transform by Theorem C and Lemma T7.

Calculus status: Exact linear at the stability condition; carried to the continuum by Lemma D7, and the input was algebraic, so the differential content is entirely supplied by the transform.

36. Magnetostrictive Strain

Origin (LA edition).

$$\lambda = \frac{3}{2}\lambda_s \left(\frac{M}{M_s}\right)^2$$

Input (8S edition).

$$E_{36}^{8S}(z) = z^\top \mathcal{G}_8 z\Big|_{\text{node } 36}, \qquad \|z_{36}(u) - z_j(u)\|_{\mathcal{G}_8} = \delta_{36j}^{(8)} = \sqrt{(c_{36} - c_j)^\top G_5 (c_{36} - c_j) + (r_{36} - r_j)^2}$$

Calculus image.

$$\mathcal{E}_{36}[\phi] = \tfrac{1}{2}\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc, \qquad \delta\mathcal{E}_{36} = 0 \iff \mathcal{L}_\star \phi = 0, \qquad ds_{36}^2 = dc^\top (G_5 + \nabla r\, \nabla r^\top) dc + r_{36}^2\, d\Omega_3^2$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T3**, so the calculus image is of class **D3** (Dirichlet energy and the cone metric). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Dirichlet energy and the cone metric of Lemma D3, instantiated at the point $c_{36} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D3 needs **(R3)**, $r \in C^1$ with $r > 0$, for the bundle to be a smooth manifold; at $r = 0$ the fibre collapses and ds^2 degenerates to the base metric. The integration by parts behind Θ-C(1) needs either a Dirichlet or a Neumann boundary condition; on the penteract the latter is the one the kernel limit actually imposes.

(Q4) Derivation. Two derivations meet here. The energy: integrate $\langle \mathcal{L}_\star \phi, \phi \rangle_{p^2}$ by parts, the weight p^2 cancelling against $1/p^2$, to obtain $\frac{1}{2}\int \|\nabla\phi\|^2_{G_5^{-1}} p^2$. The metric: differentiate the embedding $\Phi(c,u) = (c, r(c)u)$, use $u \cdot du = \frac{1}{2} d(\|u\|^2) = 0$ to kill the cross term and $\|u\| = 1$ to normalise the rest, and collect $(\nabla r \cdot dc)^2 = dc^\top \nabla r \nabla r^\top dc$.

(Q5) Retraction through the tower. Replacing ∇ by lattice differences returns the 8S quadratic form $x^\top \mathcal{L}_{8S} x$, and freezing r to a constant collapses ds^2 to the chord $\delta^{(8)}$. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(36)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{36}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Quadratic in magnetisation.

Structure: Like electrostriction, quadratic and hence with no linear coupling block. Biasing about M_0 gives an effective linear piezomagnetic coefficient, which is how magnetostrictive actuators are operated.

Reduction: Linearisation about a bias magnetisation.

Status: Linearised.

8S status: Linearised; the 8S image inherits the same approximation status (Theorem C, Lemma T3).

Calculus status: Linearised; carried to the continuum by Lemma D3, and the input was algebraic, so the differential content is entirely supplied by the transform.

37. Villari (Inverse Magnetostrictive) Effect

Origin (LA edition).

$$\Delta\mu = f(\sigma)$$

Input (8S edition).

$$\left.\frac{\partial \dot{\theta}}{\partial \theta}\right|_{\theta^\star} = -\mathcal{L}_{8S}, \qquad \left.\frac{\partial \Phi}{\partial p}\right|_{p^\star} = \Sigma'(2J_2 W + 4J_3 \widehat{H}_{p^\star}), \qquad \Sigma' = \operatorname{diag}(p^\star_{37}(1 - p^\star_{37}))$$

Calculus image.

$$\mathcal{N}_{37}(x_0 + \xi) = \mathcal{N}_{37}(x_0) + D\mathcal{N}_{37}[x_0]\,\xi + R_{37}[\xi], \qquad \big\|R_{37}[\xi]\big\| \leq \tfrac{1}{2} \sup_{\|\zeta\| \leq \|\xi\|} \big\|D^2 \mathcal{N}_{37}[x_0 + \zeta]\big\| \, \|\xi\|^2$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T13**, so the calculus image is of class **D13** (Fréchet derivative with a quantified remainder). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fréchet derivative with a quantified remainder of Lemma D13, instantiated at the point $c_{37} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D13 needs $\mathcal{N}$ twice Fréchet differentiable on a neighbourhood of x_0 for Taylor's theorem with integral remainder, and needs condition **(C)**, $\frac{1}{2}|J_2|\Delta_W + |J_3|\Delta_{\hat{H}} < 1$, for the point x_0 to exist and be unique — by Θ-H(2) that is exactly strict convexity of the free energy. Beyond (C) the expansion point is not unique and the remainder bound is void.

(Q4) Derivation. Expand $\mathcal{N}$ about the fixed point by Taylor's theorem with integral remainder, $R[\xi] = \int_0^1 (1-t) D^2\mathcal{N}[x_0 + t\xi][\xi, \xi]\, dt$, and bound the integral by the supremum of the second derivative. The phase block of $D\mathcal{N}$ is $-\mathcal{L}_\star$ by Θ-G; the gate block is $\nabla^2 \mathcal{F}$, whose positive definiteness under (C) is Θ-H(2) and which certifies that x_0 is the unique expansion point.

(Q5) Retraction through the tower. Discretising both Jacobian blocks returns the 8S two-block Jacobian, $-\mathcal{L}_{8S}$ and $\Sigma'(2J_2W+4J_3\widehat{H})$; the remainder term has no 8S counterpart and is new information supplied by the calculus edition. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(37)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{37}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The reciprocal off-diagonal block to magnetostriction.

Structure: Reciprocity requires the two off-diagonal blocks to be transposes when both are linearised about the same bias — an Onsager relation. Used for stress sensing.

Reduction: The effect is the transposed coupling block.

Status: Linearised.

8S status: Linearised about the 8S fixed point $(p^\star, \theta^\star)$ (Lemma T13); the image is an approximation exactly as the input is.

Calculus status: Linearised, now with a quantified remainder: $\|R\| \leq \frac{1}{2}\sup\|D^2\mathcal{N}\|\,\|\xi\|^2$ (Lemma D13). The calculus edition supplies a modulus of validity the 8S edition could only assert.

38. Magnetohydrodynamic Induction Equation

Origin (LA edition).

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) + \eta \nabla^2 \mathbf{B}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(38)} = \mu_k^{(38)}\phi_k^{(38)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(38)} \leq \mu_2^{(38)} \leq \cdots \leq \mu_N^{(38)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(38)}) = \mu_k^{(38)}\phi_k^{(38)}, \qquad \mu_2^{(38)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is second order — it already carries a Laplacian, a second derivative, or a divergence. The input already carries a second-order operator. The transform therefore does not manufacture the ellipticity of the image; it identifies the input's own second-order part with the principal symbol $-\frac{1}{2}\xi^\top G_5^{-1}\xi$ of $\mathcal{L}_\star$, and only the coupling and the gate are new.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{38} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(38)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{38}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Advection plus diffusion of a vector field.

Structure: Linear in $\mathbf{B}$ for a prescribed $\mathbf{v}$, so the kinematic dynamo problem is an eigenvalue problem for the induction operator — growth occurs when an eigenvalue has positive real part. The full MHD problem is nonlinear through the Lorentz force feedback.

Reduction: The equation is linear in $\mathbf{B}$ for given $\mathbf{v}$.

Status: Exact linear in the kinematic problem.

8S status: Exact linear in the kinematic problem; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear in the kinematic problem; carried to the continuum by Lemma D2, and the input already carried a second-order operator, identified with the principal symbol of $\mathcal{L}_\star$.

Also in: Plasma Physics, Geophysics.

39. Lorentz Force Density

Origin (LA edition).

$$\mathbf{f} = \mathbf{J} \times \mathbf{B} = [\mathbf{J}]_\times \mathbf{B}$$

Input (8S edition).

$$y_{39}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{39})\Big] z(u), \qquad z_{39}(u) = \begin{bmatrix} c_{39} \\ r_{39} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{39})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{39}\, \psi_{39}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{39} = -\mathcal{L}_\star \psi_{39}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{39} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the

Gaussian, use the vanishing of the odd moments and $\int w\,uu^{\top}du = \sigma^2\kappa\,G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W\Gamma)\otimes\mathsf{B}(\mathsf{A}_{39})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma\to I$, $W\to I$, $\mathcal{R}\to 0$, $K_2,K_3,J_2,J_3\to 0$ and $\Pi\to[\,I_3\;0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S}-\nu(\varepsilon_{\mathrm{ind}}-\eta^{(39)}_{\mathrm{ind}})]$ vanishes with $\nu\ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{39}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A skew matrix acting on the magnetic field.

Structure: Skew-symmetry means the force is perpendicular to both current and field, so it does no work on the current carriers. Decomposes into magnetic pressure and tension via the Maxwell stress tensor.

Reduction: The cross product is a skew matrix.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

40. Magnetic Reynolds Number

Origin (LA edition).

$$\mathrm{Rm}=\frac{UL}{\eta}$$

Input (8S edition).

$$\left(g^{(5)}_{40j},\ \delta^{(8)}_{40j},\ g^{(3)}_{40j},\ g^{(\mathrm{J})}_{40j}\right)\ \longmapsto\ \mathcal{R}^{8S}_{40j},\qquad g^{(3)}_{40j}\le g^{(5)}_{40j}$$

Calculus image.

$$d_g(z_{40},z_j)=\inf_{\Gamma}\int_0^1\|\dot\Gamma\|_{ds}\,dt,\qquad \|\nabla d_g\|_{ds^{-1}}=1\ \text{a.e.},\qquad \delta^{(8)}_{40j}\le d_g(z_{40},z_j),\qquad g^{(3)}_{40j}\le g^{(5)}_{40j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{40}\in\mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)}\le g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)}\le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{40j}$ and the gap quadruple is recovered. Composing with the 8S retraction of

Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(40)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{40}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The ratio of advection to diffusion in the induction operator.

Structure: The relative weight of the two terms in the induction equation. Large Rm means the advective (non-normal) part dominates, giving field line freezing and transient amplification; small Rm means the symmetric diffusive part dominates and the field decays.

Reduction: The number weights two operator terms.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input was algebraic, so the differential content is entirely supplied by the transform.

41. Hartmann Number

Origin (LA edition).

$$\mathrm{Ha} = BL\sqrt{\frac{\sigma}{\mu}}$$

Input (8S edition).

$$y_{41}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{41})\Big] z(u), \qquad z_{41}(u) = \begin{bmatrix} c_{41} \\ r_{41} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{41})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{41}\, \psi_{41}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{41} = -\mathcal{L}_\star \psi_{41}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{41} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{41})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of

that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(41)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{41}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The ratio of Lorentz to viscous forces.

Structure: Large Ha produces the Hartmann boundary layer of thickness L/Ha, a boundary-layer eigenmode of the coupled operator. The flow profile flattens because the magnetic term dominates the viscous one in the interior.

Reduction: The number weights two terms of a linear operator.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

Chemo-Mechanics and Electrokinetics

42. Chemical Potential with Stress

Origin (LA edition).

$$\mu = \mu_0 + RT \ln a - \Omega \, \sigma_h$$

Input (8S edition).

$$y_{42}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{42})\Big] z(u), \qquad z_{42}(u) = \begin{bmatrix} c_{42} \\ r_{42} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{42})(c) = \int_{\mathcal{C}} \gamma(c) \, W_\sigma(c, c') \, \gamma(c') \, \mathsf{A}_{42} \, \psi_{42}(c') \, p(c') \, dc', \qquad \partial_\tau \psi_{42} = -\mathcal{L}_\star \psi_{42}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot \,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{42} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p \, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w \, u u^\top du = \sigma^2 \kappa \, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{42})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of

that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(42)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{42}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A linear coupling of hydrostatic stress into the chemical potential.

Structure: $\sigma_h = \frac{1}{3}\operatorname{tr}\sigma$, so only the spherical part couples — the deviator does not affect diffusion at this order. The coupling coefficient Ω is the partial molar volume.

Reduction: The coupling is linear in a trace.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

43. Stress-Assisted Diffusion Flux

Origin (LA edition).

$$\mathbf{J} = -D\nabla c + \frac{Dc\Omega}{RT}\nabla\sigma_h$$

Input (8S edition).

$$y_{43}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{43})\Big] z(u), \qquad z_{43}(u) = \begin{bmatrix} c_{43} \\ r_{43} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{43})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{43}\, \psi_{43}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{43} = -\mathcal{L}_\star \psi_{43}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\,\cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{43} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{43})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of

that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(43)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{43}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Two driving gradients with a matrix diffusivity.

Structure: The flux is a linear map of the combined driving force $\nabla\mu$; writing it as $\mathbf{J} = -L\nabla\mu$ with $L \succeq 0$ makes the Onsager structure explicit. The two terms are the diagonal and off-diagonal contributions.

Reduction: The flux is a matrix times a generalised gradient.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input already carried a first-order operator, which the transform leaves in place.

44. Diffusion-Induced Stress

Origin (LA edition).

$$\sigma = -\frac{E\Omega}{3(1-\nu)}(c - \bar{c})I$$

Input (8S edition).

$$y_{44}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{44})\Big] z(u), \qquad z_{44}(u) = \begin{bmatrix} c_{44} \\ r_{44} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{44})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{44}\, \psi_{44}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{44} = -\mathcal{L}_\star \psi_{44}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla\, \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{44} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{44})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(44)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{44}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An eigenstrain from composition, structurally identical to thermal strain.

Structure: Concentration plays exactly the role of temperature in the Duhamel–Neumann law, so the chemo-mechanical block matrix has the same form as the thermomechanical one. The analogy is exact at the level of the matrix structure.

Reduction: Composition acts as a thermal-like eigenstrain.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

45. Swelling Pressure (Flory–Rehner)

Origin (LA edition).

$$\Pi = -\frac{RT}{V_1}[\ln(1-\phi) + \phi + \chi\phi^2] - \frac{RT}{V_1} N(\phi^{1/3} - \phi/2)$$

Input (8S edition).

$$\left.\frac{\partial \dot{\theta}}{\partial \theta}\right|_{\theta^\star} = -\mathcal{L}_{8S}, \qquad \left.\frac{\partial \Phi}{\partial p}\right|_{p^\star} = \Sigma'(2J_2 W + 4J_3 \widehat{H}_{p^\star}), \qquad \Sigma' = \operatorname{diag}(p_{45}^\star(1 - p_{45}^\star))$$

Calculus image.

$$\mathcal{N}_{45}(x_0 + \xi) = \mathcal{N}_{45}(x_0) + D\mathcal{N}_{45}[x_0]\,\xi + R_{45}[\xi], \qquad \big\|R_{45}[\xi]\big\| \leq \tfrac{1}{2} \sup_{\|\zeta\| \leq \|\xi\|} \big\|D^2 \mathcal{N}_{45}[x_0 + \zeta]\big\|\, \|\xi\|^2$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T13**, so the calculus image is of class **D13** (Fréchet derivative with a quantified remainder). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fréchet derivative with a quantified remainder of Lemma D13, instantiated at the point $c_{45} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D13 needs $\mathcal{N}$ twice Fréchet differentiable on a neighbourhood of x_0 for Taylor's theorem with integral remainder, and needs condition **(C)**, $\frac{1}{2}|J_2|\Delta_W + |J_3|\Delta_{\hat{H}} < 1$, for the point x_0 to exist and be unique — by Θ-H(2) that is exactly strict convexity of the free energy. Beyond (C) the expansion point is not unique and the remainder bound is void.

(Q4) Derivation. Expand $\mathcal{N}$ about the fixed point by Taylor's theorem with integral remainder, $R[\xi] = \int_0^1 (1-t) D^2\mathcal{N}[x_0 + t\xi][\xi,\xi]\, dt$, and bound the integral by the supremum of the second derivative. The phase block of $D\mathcal{N}$ is $-\mathcal{L}_\star$ by Θ-G; the gate block is $\nabla^2 \mathcal{F}$, whose positive definiteness under (C) is Θ-H(2) and which certifies that x_0 is the unique expansion point.

(Q5) Retraction through the tower. Discretising both Jacobian blocks returns the 8S two-block Jacobian, $-\mathcal{L}_{8S}$ and $\Sigma'(2J_2 W + 4J_3 \widehat{H})$; the remainder term has no 8S counterpart and is new information supplied by the calculus edition. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(45)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{45}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A nonlinear function of the polymer fraction.

Structure: Genuinely nonlinear in ϕ. Linearising about the equilibrium swelling ratio gives an effective osmotic modulus, which is what enters the linearised poroelastic block matrix.

Reduction: Linearisation about equilibrium gives a modulus.

Status: Linearised.

8S status: Linearised about the 8S fixed point $(p^\star, \theta^\star)$ (Lemma T13); the image is an approximation exactly as the input is.

Calculus status: Linearised, now with a quantified remainder: $\|R\| \leq \frac{1}{2} \sup \|D^2\mathcal{N}\| \, \|\xi\|^2$ (Lemma D13). The calculus edition supplies a modulus of validity the 8S edition could only assert.

46. Poroelastic Swelling Time

Origin (LA edition).

$$\tau = \frac{L^2}{D_{\text{eff}}}$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(46)} = \mu_k^{(46)} \phi_k^{(46)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(46)} \leq \mu_2^{(46)} \leq \cdots \leq \mu_N^{(46)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \phi_k^{(46)}) = \mu_k^{(46)} \phi_k^{(46)}, \qquad \mu_2^{(46)} = \min_{\substack{\phi \neq 0 \\ \int \phi \, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla \phi\|^2_{G_5^{-1}} \, p^2 \, dc}{\int_{\mathcal{C}} \phi^2 \, p^2 \, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{46} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(46)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{46}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The reciprocal of the smallest nonzero diffusion eigenvalue.

Structure: Diffusive scaling means τ is set by λ_1 of the discretised Laplacian on the domain, which scales as L^{-2}. The relaxation spectrum is the full eigenvalue set.

Reduction: The time constant is a reciprocal eigenvalue.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

47. Nernst–Planck Equation

Origin (LA edition).

$$\mathbf{J}_i = -D_i \nabla c_i - \frac{z_i F}{RT} D_i c_i \nabla \phi + c_i \mathbf{v}$$

Input (8S edition).

$$\left.\frac{\partial \dot{\theta}}{\partial \theta}\right|_{\theta^\star} = -\mathcal{L}_{8S}, \qquad \left.\frac{\partial \Phi}{\partial p}\right|_{p^\star} = \Sigma'(2J_2 W + 4J_3 \widehat{H}_{p^\star}), \qquad \Sigma' = \operatorname{diag}(p_{47}^\star(1 - p_{47}^\star))$$

Calculus image.

$$\mathcal{N}_{47}(x_0 + \xi) = \mathcal{N}_{47}(x_0) + D\mathcal{N}_{47}[x_0]\,\xi + R_{47}[\xi], \qquad \big\|R_{47}[\xi]\big\| \leq \tfrac{1}{2} \sup_{\|\zeta\| \leq \|\xi\|} \big\|D^2 \mathcal{N}_{47}[x_0 + \zeta]\big\| \, \|\xi\|^2$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T13**, so the calculus image is of class **D13** (Fréchet derivative with a quantified remainder). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Fréchet derivative with a quantified remainder of Lemma D13, instantiated at the point $c_{47} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D13 needs $\mathcal{N}$ twice Fréchet differentiable on a neighbourhood of x_0 for Taylor's theorem with integral remainder, and needs condition **(C)**, $\frac{1}{2}|J_2|\Delta_W + |J_3|\Delta_{\hat{H}} < 1$, for the point x_0 to exist and be unique — by Θ-H(2) that is exactly strict convexity of the free energy. Beyond (C) the expansion point is not unique and the remainder bound is void.

(Q4) Derivation. Expand $\mathcal{N}$ about the fixed point by Taylor's theorem with integral remainder, $R[\xi] = \int_0^1 (1-t) D^2 \mathcal{N}[x_0 + t\xi][\xi, \xi]\, dt$, and bound the integral by the supremum of the second derivative. The phase block of $D\mathcal{N}$ is $-\mathcal{L}_\star$ by Θ-G; the gate block is $\nabla^2 \mathcal{F}$, whose positive definiteness under (C) is Θ-H(2) and which certifies that x_0 is the unique expansion point.

(Q5) Retraction through the tower. Discretising both Jacobian blocks returns the 8S two-block Jacobian, $-\mathcal{L}_{8S}$ and $\Sigma'(2J_2 W + 4J_3 \widehat{H})$; the remainder term has no 8S counterpart and is new information supplied by the calculus edition. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(47)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{47}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Diffusion, migration, and advection combined linearly.

Structure: Linear in the fluxes given the fields, but coupled to Poisson's equation for ϕ — the resulting Poisson–Nernst–Planck system is nonlinear through the $c_i \nabla\phi$ product. Linearising about equilibrium gives the Debye–Hückel theory.

Reduction: Linear in the fluxes; nonlinear through the field coupling.

Status: Linearised.

8S status: Linearised about the 8S fixed point $(p^\star, \theta^\star)$ (Lemma T13); the image is an approximation exactly as the input is.

Calculus status: Linearised, now with a quantified remainder: $\|R\| \leq \frac{1}{2} \sup \|D^2\mathcal{N}\| \, \|\xi\|^2$ (Lemma D13). The calculus edition supplies a modulus of validity the 8S edition could only assert.

Also in: Physical Chemistry.

48. Poisson–Boltzmann Equation

Origin (LA edition).

$$\nabla^2\phi = -\frac{1}{\epsilon} \sum_i z_i e c_i^\infty e^{-z_i e\phi/kT}$$

Input (8S edition).

$$\mathcal{L}_{8S}\, \phi_k^{(48)} = \mu_k^{(48)} \phi_k^{(48)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(48)} \leq \mu_2^{(48)} \leq \cdots \leq \mu_N^{(48)}$$

Calculus image.

$$-\frac{1}{2p^2} \nabla\cdot(p^2 G_5^{-1} \nabla\phi_k^{(48)}) = \mu_k^{(48)} \phi_k^{(48)}, \qquad \mu_2^{(48)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is second order — it already carries a Laplacian, a second derivative, or a divergence. The input already carries a second-order operator. The transform therefore does not manufacture the ellipticity of the image; it identifies the input's own second-order part with the principal symbol $-\frac{1}{2}\xi^\top G_5^{-1}\xi$ of $\mathcal{L}_\star$, and only the coupling and the gate are new.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{48} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing

with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(48)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{48}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A nonlinear Poisson equation.

Structure: The exponential makes it nonlinear; linearising for $|ze\phi| \ll kT$ gives the Debye–Hückel equation $\nabla^2\phi = \kappa^2\phi$ — a Helmholtz operator whose eigenvalue κ^2 sets the screening length.

Reduction: Debye–Hückel linearisation gives a Helmholtz operator.

Status: Linearised.

8S status: Linearised; the 8S image inherits the same approximation status (Theorem C, Lemma T2).

Calculus status: Linearised; carried to the continuum by Lemma D2, and the input already carried a second-order operator, identified with the principal symbol of $\mathcal{L}_\star$.

49. Debye Length

Origin (LA edition).

$$\lambda_D = \sqrt{\frac{\epsilon k T}{2 n_0 z^2 e^2}}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(49)} = \mu_k^{(49)}\phi_k^{(49)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(49)} \leq \mu_2^{(49)} \leq \cdots \leq \mu_N^{(49)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(49)}) = \mu_k^{(49)}\phi_k^{(49)}, \qquad \mu_2^{(49)} = \min_{\substack{\phi \neq 0 \\ \int \phi\, p^2 dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{49} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing

with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(49)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{49}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The reciprocal of the Helmholtz operator's eigenvalue.

Structure: $\lambda_D = 1/\kappa$ where κ^2 is the coefficient in the linearised Poisson–Boltzmann operator. Screening is exponential decay at the rate set by that eigenvalue.

Reduction: The length is a reciprocal operator eigenvalue.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Plasma Physics.

50. Electroosmotic Velocity (Smoluchowski)

Origin (LA edition).

$$u_{eo} = -\frac{\epsilon \zeta E}{\mu}$$

Input (8S edition).

$$y_{50}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{50})\Big] z(u), \qquad z_{50}(u) = \begin{bmatrix} c_{50} \\ r_{50} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{50})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{50}\, \psi_{50}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{50} = -\mathcal{L}_\star \psi_{50}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{50} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{50})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0 \,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(50)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{50}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A linear response of velocity to applied field.

Structure: Plug-like flow, linear in E — one entry of the electrokinetic coupling matrix. The mobility $-\epsilon\zeta/\mu$ is the off-diagonal Onsager coefficient linking electrical and hydrodynamic transport.

Reduction: The velocity is a linear response coefficient.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

51. Electrophoretic Mobility

Origin (LA edition).

$$\mu_e = \frac{\epsilon\zeta}{\mu}$$

Input (8S edition).

$$y_{51}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{51})\Big] z(u), \qquad z_{51}(u) = \begin{bmatrix} c_{51} \\ r_{51} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_\sigma \psi_{51})(c) = \int_{\mathcal{C}} \gamma(c)\, W_\sigma(c, c')\, \gamma(c')\, \mathsf{A}_{51}\, \psi_{51}(c')\, p(c')\, dc', \qquad \partial_\tau \psi_{51} = -\mathcal{L}_\star \psi_{51}, \qquad \mathcal{L}_\star = -\frac{1}{2p^2} \nabla \cdot (p^2 G_5^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{51} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_\sigma$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^4 regularity for the $O(\sigma^2)$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_j (\cdot)_j$ by $\int_{\mathcal{C}} (\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^\top du = \sigma^2 \kappa\, G_5^{-1}$, and divide by the degree. What survives at order σ^2 is $-\mathcal{L}_\star$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_M$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{51})$ with $O(\ell^2)$ quadrature error. Composing with the 8S retraction of Theorem J of

that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta^{(51)}_{\text{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{51}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The particle-frame counterpart of electroosmosis.

Structure: Same magnitude, opposite sign — the two effects are the same off-diagonal coefficient viewed from two frames. Their equality is a reciprocity statement.

Reduction: The mobility is the same coupling coefficient.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input was algebraic, so the differential content is entirely supplied by the transform.

52. Streaming Potential

Origin (LA edition).

$$\frac{\Delta\phi}{\Delta p} = \frac{\epsilon\zeta}{\mu\sigma}$$

Input (8S edition).

$$\left(g^{(5)}_{52j},\ \delta^{(8)}_{52j},\ g^{(3)}_{52j},\ g^{(\mathrm{J})}_{52j}\right) \;\longmapsto\; \mathcal{R}^{8S}_{52j}, \qquad g^{(3)}_{52j} \leq g^{(5)}_{52j}$$

Calculus image.

$$d_g(z_{52}, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{52j} \leq d_g(z_{52}, z_j), \qquad g^{(3)}_{52j} \leq g^{(5)}_{52j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{52} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \leq g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \leq d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{52j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \; 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(52)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{52}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The reciprocal off-diagonal coefficient to electroosmosis.

Structure: Onsager reciprocity requires the pressure-driven current coefficient to equal the field-driven flow coefficient — the electrokinetic matrix $\left(\begin{smallmatrix} L_{11} & L_{12} \\ L_{21} & L_{22} \end{smallmatrix}\right)$ is symmetric. Streaming potential and electroosmosis are L_{21} and L_{12}.

Reduction: The effect is the transposed coupling coefficient.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input was algebraic, so the differential content is entirely supplied by the transform.

53. Onsager Reciprocal Relations

Origin (LA edition).

$$L_{ij} = L_{ji}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(53)} = \mu_k^{(53)}\phi_k^{(53)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(53)} \leq \mu_2^{(53)} \leq \cdots \leq \mu_N^{(53)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(53)}) = \mu_k^{(53)}\phi_k^{(53)}, \qquad \mu_2^{(53)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2\, dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{53} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(53)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{53}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Symmetry of the transport coefficient matrix.

Structure: The central structural theorem of coupled transport: the phenomenological matrix is symmetric, a consequence of microscopic time-reversibility. Symmetry means the flux–force system is variational, guarantees real eigenvalues, and halves the number of independent coefficients. In a magnetic field the relation becomes $L_{ij}(\mathbf{B}) = L_{ji}(-\mathbf{B})$.

Reduction: Reciprocity is matrix symmetry.

Status: Exact linear. The organising principle of this chapter.

8S status: Exact linear. The organising principle of this chapter; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear. The organising principle of this chapter; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

Also in: Thermodynamics, Statistical Mechanics.

54. Thermoelectric Coupled Transport

Origin (LA edition).

$$\begin{pmatrix} \mathbf{J} \\ \mathbf{q} \end{pmatrix} = \begin{pmatrix} \sigma & \sigma S \\ \sigma S T & \kappa + \sigma S^2 T \end{pmatrix} \begin{pmatrix} \mathbf{E} \\ -\nabla T \end{pmatrix}$$

Input (8S edition).

$$W_{54j} = A_{54j} \exp\left[-\frac{(c_{54} - c_j)^\top G_5 (c_{54} - c_j)}{2\sigma_c^2} - \frac{d_{\mathrm{J}}(54, j)^2}{2\sigma_{\mathrm{J}}^2} \right], \qquad W \succeq 0$$

Calculus image.

$$\partial_t k_t = \nabla \cdot (G_5^{-1} \nabla k_t), \quad k_0 = \delta_0; \qquad W_\sigma\big|_{\text{node } 54} = e^{t\mathcal{A}_G}, \quad t = \tfrac{\sigma_c^2}{2}; \qquad \widehat{k_t}(\xi) = e^{-t\,\xi^\top G_5^{-1} \xi}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T8**, so the calculus image is of class **D8** (heat semigroup and its symbol). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the heat semigroup and its symbol of Lemma D8, instantiated at the point $c_{54} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D8 needs $G_5 \succ 0$ for the Gaussian to be integrable and for $G_5^{1/2}$ to exist. No hypothesis on the tag factor is required for the heat-kernel identification, since that factor is a smooth positive multiplier; where it is retained, it is absorbed into the sampling density p.

(Q4) Derivation. Substitute $y = G_5^{1/2} x$, under which $\nabla \cdot (G_5^{-1} \nabla \cdot)$ becomes the ordinary Laplacian and the coupling kernel becomes the standard Gaussian. Differentiating $t^{-5/2} e^{-\|y\|^2/4t}$ in t and in y verifies the heat equation directly. Matching exponents $\|x\|_{G_5}^2 / 2\sigma_c^2 = \|x\|_{G_5}^2 / 4t$ identifies $t = \sigma_c^2/2$. The symbol follows from the Gaussian Fourier integral $\int e^{-x^\top A x} e^{-i\xi \cdot x} dx = \pi^{5/2} \det(A)^{-1/2} e^{-\xi^\top A^{-1} \xi / 4}$.

(Q5) Retraction through the tower. Sampling the heat kernel at lattice separations returns W_{ij} exactly, since the 8S kernel *is* k_t evaluated at $c_i - c_j$ with $t = \sigma_c^2/2$. Composing with the 8S retraction of Theorem J of that edition,

which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta^{(54)}_{\mathrm{ind}})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{54}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A symmetric 2×2 block transport matrix.

Structure: Seebeck and Peltier effects are the two off-diagonal entries, equal by Onsager after the factor T — the Kelvin relation $\Pi = ST$. Positive semidefiniteness of the matrix is the second law.

Reduction: Coupling is an off-diagonal transport block.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T8.

Calculus status: Exact linear; carried to the continuum by Lemma D8, and the input already carried a first-order operator, which the transform leaves in place.

55. Thermoelectric Figure of Merit

Origin (LA edition).

$$ZT = \frac{S^2 \sigma T}{\kappa}$$

Input (8S edition).

$$\left(g^{(5)}_{55j},\ \delta^{(8)}_{55j},\ g^{(3)}_{55j},\ g^{(\mathrm{J})}_{55j}\right) \;\longmapsto\; \mathcal{R}^{8S}_{55j}, \qquad g^{(3)}_{55j} \leq g^{(5)}_{55j}$$

Calculus image.

$$d_g(z_{55}, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{55j} \leq d_g(z_{55}, z_j), \qquad g^{(3)}_{55j} \leq g^{(5)}_{55j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{55} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \leq g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \leq d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{55j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(55)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{55}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: A normalised measure of off-diagonal dominance.

Structure: Exactly the ratio of the squared off-diagonal to the product of diagonals in the transport matrix — the same structural quantity as the piezoelectric coupling factor. Efficiency increases monotonically with ZT, and $ZT \to \infty$ would reach Carnot.

Reduction: The figure of merit measures block off-diagonal dominance.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input was algebraic, so the differential content is entirely supplied by the transform.

56. Soret (Thermodiffusion) Flux

Origin (LA edition).

$$\mathbf{J} = -D\nabla c - D_T c(1-c)\nabla T$$

Input (8S edition).

$$\left(g_{56j}^{(5)},\ \delta_{56j}^{(8)},\ g_{56j}^{(3)},\ g_{56j}^{(\mathrm{J})}\right) \ \longmapsto\ \mathcal{R}_{56j}^{8S}, \qquad g_{56j}^{(3)} \leq g_{56j}^{(5)}$$

Calculus image.

$$d_g(z_{56}, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta_{56j}^{(8)} \leq d_g(z_{56}, z_j), \qquad g_{56j}^{(3)} \leq g_{56j}^{(5)}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{56} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \leq g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \leq d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta_{56j}^{(8)}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(56)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{56}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: An off-diagonal coefficient coupling temperature gradient to mass flux.

Structure: The L_{12} entry of the mass–heat transport matrix. Its sign varies with system, and it is the mechanism behind thermophoresis and isotope separation columns.

Reduction: The effect is an off-diagonal transport coefficient.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input already carried a first-order operator, which the transform leaves in place.

57. Dufour Effect

Origin (LA edition).

$$\mathbf{q} = -\kappa \nabla T - \rho D_{F} \nabla c$$

Input (8S edition).

$$y_{57}(\tau, u) = \Pi(\tau)\Big[(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{57})\Big] z(u), \qquad z_{57}(u) = \begin{bmatrix} c_{57} \\ r_{57} u \end{bmatrix}$$

Calculus image.

$$(\mathcal{K}_{\sigma}\psi_{57})(c) = \int_{\mathcal{C}} \gamma(c)\, W_{\sigma}(c, c')\, \gamma(c')\, \mathsf{A}_{57}\, \psi_{57}(c')\, p(c')\, dc', \qquad \partial_{\tau}\psi_{57} = -\mathcal{L}_{\star}\psi_{57}, \qquad \mathcal{L}_{\star} = -\frac{1}{2p^{2}} \nabla \cdot (p^{2} G_{5}^{-1} \nabla \cdot\,)$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T1**, so the calculus image is of class **D1** (Fredholm integral operator and its generator). The originating equation is first order — it already carries a derivative or an integral. The input already carries a first-order differential or integral operator. On that part the transform acts as the identity — a derivative is already a calculus object — and only the coupling and the gate are taken to the continuum. The regularity demanded of the input is correspondingly one order lower.

(Q2) Image. The displayed relation is the Fredholm integral operator and its generator of Lemma D1, instantiated at the point $c_{57} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D1 needs the kernel continuous and square-integrable on $\mathcal{C} \times \mathcal{C}$, which the Gaussian form supplies unconditionally, so $\mathcal{K}_{\sigma}$ is Hilbert–Schmidt and hence compact. The generator statement needs **(R1)** and **(R2)**: C^{4} regularity for the $O(\sigma^{2})$ rate of Θ-G, and interior evaluation, since within $O(\sigma)$ of $\partial\mathcal{C}$ the expansion picks up the Neumann boundary term of Remark Θ-G″.

(Q4) Derivation. The 8S image is a Kronecker product $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A})$ acting by a sum over nodes. Definition 2.2 replaces $\sum_{j}(\cdot)_{j}$ by $\int_{\mathcal{C}}(\cdot)\, p\, dc'$, turning the matrix into the displayed Fredholm operator of the second kind. Its bandwidth limit is computed by Θ-G: Taylor-expand ψ and p to second order about c, integrate against the Gaussian, use the vanishing of the odd moments and $\int w\, uu^{\top} du = \sigma^{2} \kappa\, G_{5}^{-1}$, and divide by the degree. What survives at order σ^{2} is $-\mathcal{L}_{\star}$.

(Q5) Retraction through the tower. Sampling the integral operator on $\mathcal{C}_{M}$ by the Riemann rule of Θ-J(1) returns the finite sum $(\Gamma W \Gamma) \otimes \mathsf{B}(\mathsf{A}_{57})$ with $O(\ell^{2})$ quadrature error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_{2}, K_{3}, J_{2}, J_{3} \to 0$ and $\Pi \to [\, I_{3}\ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(57)})]$ vanishes with $\nu \geq 0$ and complementary slackness holds. By Θ-A(3) the gradient of the

novelty is $2w_{57}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The reciprocal of the Soret effect.

Structure: L_{21} to Soret's L_{12}, equal by Onsager reciprocity. Usually negligible in liquids but significant in gases, where the two coefficients are comparable.

Reduction: The effect is the transposed coefficient.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T1.

Calculus status: Exact linear; carried to the continuum by Lemma D1, and the input already carried a first-order operator, which the transform leaves in place.

58. Coupled Field Problem Monolithic Form

Origin (LA edition).

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \end{pmatrix}$$

Input (8S edition).

$$\left(g^{(5)}_{58j},\ \delta^{(8)}_{58j},\ g^{(3)}_{58j},\ g^{(\mathrm{J})}_{58j}\right) \ \longmapsto\ \mathcal{R}^{8S}_{58j}, \qquad g^{(3)}_{58j} \le g^{(5)}_{58j}$$

Calculus image.

$$d_g(z_{58}, z_j) = \inf_{\Gamma} \int_0^1 \|\dot\Gamma\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta^{(8)}_{58j} \le d_g(z_{58}, z_j), \qquad g^{(3)}_{58j} \le g^{(5)}_{58j}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{58} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \le g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta^{(8)}_{58j}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta^{(58)}_{\mathrm{ind}})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{58}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: The general two-field block system.

Structure: The whole chapter in one equation. Symmetric off-diagonals ($A_{12} = A_{21}^\top$) give a variational structure and permit symmetric solvers; unsymmetric ones require GMRES. Block preconditioners built from the Schur complement $A_{22} - A_{21}A_{11}^{-1}A_{12}$ are the standard approach.

Reduction: All coupled problems have this form.

Status: Exact linear. The organising form of the field.

8S status: Exact linear. The organising form of the field; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear. The organising form of the field; carried to the continuum by Lemma D9, and the input was algebraic, so the differential content is entirely supplied by the transform.

59. Added Mass Instability Criterion (Partitioned FSI)

Origin (LA edition).

$$\rho_f/\rho_s \gtrsim 1 \implies \text{partitioned schemes diverge}$$

Input (8S edition).

$$\mathcal{L}_{8S}\,\phi_k^{(59)} = \mu_k^{(59)}\phi_k^{(59)}, \qquad \mathcal{L}_{8S} = K_2 L_W + 2K_3 L_{\hat{H}}, \qquad 0 = \mu_1^{(59)} \le \mu_2^{(59)} \le \cdots \le \mu_N^{(59)}$$

Calculus image.

$$-\frac{1}{2p^2}\nabla\cdot(p^2 G_5^{-1}\nabla\phi_k^{(59)}) = \mu_k^{(59)}\phi_k^{(59)}, \qquad \mu_2^{(59)} = \min_{\substack{\phi\neq 0 \\ \int \phi\, p^2\, dc = 0}} \frac{\int_{\mathcal{C}} \|\nabla\phi\|^2_{G_5^{-1}}\, p^2\, dc}{\int_{\mathcal{C}} \phi^2\, p^2\, dc}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T2**, so the calculus image is of class **D2** (Sturm–Liouville problem and Rayleigh quotient). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the Sturm–Liouville problem and Rayleigh quotient of Lemma D2, instantiated at the point $c_{59} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D2 needs **(R1)**, **(R2)** and the standing normalisation **(N)**. Discreteness of the spectrum uses compactness of $\mathcal{C}$ and the Rellich–Kondrachov embedding $H^1 \hookrightarrow L^2$; without compactness the spectrum could be continuous and μ_2 would not be attained. The min–max formula needs self-adjointness on $L^2(p^2 dc)$, which is Θ-C(2).

(Q4) Derivation. Θ-G supplies the operator by the Gaussian-moment expansion; Θ-C(1) converts it into the bilinear form by the divergence theorem, the boundary term vanishing under the Neumann condition the kernel limit imposes. Imposing the normalisation $\int \phi^2 p^2 = 1$ and taking the first variation of $\mathcal{E}$ produces the eigenvalue equation with μ as the Lagrange multiplier; Courant–Fischer then gives the displayed variational characterisation of μ_2 on the complement of the constants.

(Q5) Retraction through the tower. Discretising the Rayleigh quotient on $\mathcal{C}_M$ returns $x^\top \mathcal{L}_{8S} x / x^\top x$, whose minimiser over mean-zero x is the 8S μ_2; the gradient becomes a difference quotient with $O(\ell^2)$ error. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \;\, 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\text{ind}} - \eta_{\text{ind}}^{(59)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{59}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Spectral radius of the partitioned iteration matrix.

Structure: Partitioned solvers are block Gauss–Seidel on the coupled system, converging only when $\rho(A_{11}^{-1}A_{12}A_{22}^{-1}A_{21}) < 1$. Heavy fluids make that product's spectral radius exceed one, so no relaxation factor helps and monolithic solution is required.

Reduction: Instability is a spectral radius condition on a block product.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T2.

Calculus status: Exact linear; carried to the continuum by Lemma D2, and the input was algebraic, so the differential content is entirely supplied by the transform.

60. Coupling Strength (Dimensionless)

Origin (LA edition).

$$\Pi_c = \frac{\|A_{12}A_{21}\|}{\|A_{11}\|\|A_{22}\|}$$

Input (8S edition).

$$\left(g_{60j}^{(5)},\ \delta_{60j}^{(8)},\ g_{60j}^{(3)},\ g_{60j}^{(\mathrm{J})}\right) \longmapsto\ \mathcal{R}_{60j}^{8S}, \qquad g_{60j}^{(3)} \le g_{60j}^{(5)}$$

Calculus image.

$$d_g(z_{60}, z_j) = \inf_{\Gamma} \int_0^1 \|\dot{\Gamma}\|_{ds}\, dt, \qquad \|\nabla d_g\|_{ds^{-1}} = 1 \text{ a.e.}, \qquad \delta_{60j}^{(8)} \le d_g(z_{60}, z_j), \qquad g_{60j}^{(3)} \le g_{60j}^{(5)}$$

Proof.

(Q1) Class and differential signature. The 8S image is of class **T9**, so the calculus image is of class **D9** (geodesic distance and the eikonal equation). The originating equation is algebraic (order 0). The input is algebraic, so no part of it survives differentiation untouched; the whole of its calculus content is supplied by the transform, and the smoothing hypotheses of (R1) are doing real work.

(Q2) Image. The displayed relation is the geodesic distance and the eikonal equation of Lemma D9, instantiated at the point $c_{60} \in \mathcal{C}$ carrying this entry.

(Q3) Regularity. Lemma D9 needs **(R3)** for the metric of Θ-E to be Riemannian, and the full standing normalisation **(N)** for the ordering $g^{(3)} \le g^{(5)}$. The eikonal identity holds away from the cut locus; on the cut locus d_g is only Lipschitz and the gradient exists almost everywhere.

(Q4) Derivation. The 8S gap functions are algebraic; their calculus image is obtained by taking $\delta^{(8)}$ to be the chord of the metric of Θ-E and d_g its geodesic distance. Any curve joining two points has intrinsic length at least its ambient length, and the ambient length is at least the straight-line distance, which for a common fibre point is $\delta^{(8)}$; hence $\delta^{(8)} \le d_g$. The eikonal identity is the first-variation formula for arclength. The projected ordering is Θ-F(2), an inequality between real numbers that passes to the limit unchanged.

(Q5) Retraction through the tower. On the lattice, geodesics through a common fibre point have length equal to the chord, so d_g collapses to $\delta_{60j}^{(8)}$ and the gap quadruple is recovered. Composing with the 8S retraction of Theorem J of that edition, which sends $\Gamma \to I$, $W \to I$, $\mathcal{R} \to 0$, $K_2, K_3, J_2, J_3 \to 0$ and $\Pi \to [\, I_3 \ 0\,]$, returns the Linear Algebra equation displayed at the head of this entry. Both rungs of the tower check out. ✓

(Q6) Admission. Admission becomes a first-order condition: the entry is admitted when the constrained variation $\delta[\Delta_{8S} - \nu(\varepsilon_{\mathrm{ind}} - \eta_{\mathrm{ind}}^{(60)})]$ vanishes with $\nu \ge 0$ and complementary slackness holds. By Θ-A(3) the gradient of the novelty is $2w_{60}$, so admission asks that this entry's content have a nonzero component orthogonal to the existing frame.

Objects: Normalised off-diagonal magnitude.

Structure: The general form of k^2 and ZT — a single number measuring how far the block matrix is from block-diagonal. Small Π_c justifies staggered solution and perturbative treatment; large Π_c forces monolithic coupling. The whole practical taxonomy of coupled problems reduces to this ratio.

Reduction: Coupling strength is a normalised off-diagonal norm.

Status: Exact linear.

8S status: Exact linear; preserved under the transform by Theorem C and Lemma T9.

Calculus status: Exact linear; carried to the continuum by Lemma D9, and the input was algebraic, so the differential content is entirely supplied by the transform.